

Section B: Answer **ALL** questions. Parts marked with * involve knowledge of the extension component. Write your answers in the spaces provided.

1. Cappuccino is an Italian style coffee topped with a layer of frothy milk (Figure 1.1).



Figure 1.1

Frothy milk is made by bubbling steam through milk, which is held in a metallic jug (Figure 1.2). Steam is ejected from the steam wand of a cappuccino machine (Figure 1.3).

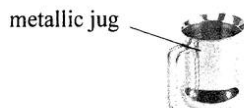


Figure 1.2



Figure 1.3

Given: specific latent heat of vaporization of water = $2.26 \times 10^6 \text{ J kg}^{-1}$
 specific heat capacity of water = $4200 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$
 specific heat capacity of steam = $2000 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$
 specific heat capacity of milk = $3900 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$

- (a) Calculate the total amount of heat released when 20 g of steam at 110°C cools to 100°C and condenses to water at 100°C . (3 marks)

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Answers written in the margins will not be marked.

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- (b) 20 g of steam at 110°C is bubbled through 200 g of milk at 15°C to make frothy milk. Using the result in (a), estimate the temperature of the frothy milk. (2 marks)

Answers written in the margins will not be marked.

- (c) Would the actual temperature of frothy milk be higher than, equal to or lower than the value found in (b)? Explain. (2 marks)

Answers written in the margins will not be marked.

Book 1 section 3.1: DSE long questions

2012 Q1 (page 3)

2012-DSE-PHY 1B-3

22